

KISHORE S

Chennai, Tamil Nadu, India | +91 76395 62179 | kishoreselvaraj146@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode \(400+ solved\)](#)

SUMMARY

Software engineer with hands-on experience shipping distributed backend services and applied-AI systems in Python and JavaScript. Strong foundation in data structures, algorithms, and object-oriented design (400+ LeetCode problems solved). Experienced with FastAPI, React, PostgreSQL, and LLM/RAG pipelines from prototype to Dockerized cloud deployment.

EDUCATION

College of Engineering Guindy, Anna University — Chennai, India

Oct 2022 – May 2027

M.Sc. (Integrated) Information Technology | CGPA: 7.1/10

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Machine Learning

EXPERIENCE

Software Developer Intern — IITM-Incubated Startup, Chennai

Jun 2025 – Nov 2025

- Developed and maintained backend booking management services using Python and FastAPI, implementing booking creation, status management, and fault-tolerant retry mechanisms.
- Designed and implemented RESTful APIs with input validation, idempotency, authentication, and structured error handling to ensure reliable and secure service communication.
- Optimized database performance by eliminating N+1 query patterns, implementing indexing strategies, query optimization, and efficient pagination to improve application scalability and responsiveness.
- Collaborated with frontend developers to integrate React applications with backend services, following Agile/Scrum methodologies, Git-based workflows, code reviews, and production deployment practices.

PROJECTS

TherapyOS — Multi-Tenant Practice-Management SaaS / Founder | FastAPI, PostgreSQL, Redis, Celery, React, Docker

- Founded and shipped a **multi-tenant SaaS platform** for pediatric therapy clinics, enforcing tenant isolation with PostgreSQL **Row-Level Security** across 16 tables and 3 role-based portals (Admin, Therapist, Parent).
- Built **AI-assisted clinical documentation** generating SOAP notes and plain-language parent summaries in a single cached LLM call, cutting note-writing time by **~70%** while holding AI cost near 3% of revenue.
- Engineered **WebSocket chat, Redis caching, and Celery** background workers for reminders and payment webhooks; partitioned the messages table across 8 shards by tenant for scale.
- Integrated **Razorpay billing and WhatsApp messaging**; containerized with Docker and drove the product from architecture to live clinic sales.

AgentOS — Multi-Agent AI Operating System | Python, LangGraph, LangChain, FAISS, MCP, FastAPI

- Engineered an autonomous multi-agent system that plans, executes, and self-reflects on tasks, orchestrating **5 specialized agents** (Planner, Research, Coding, Testing, Documentation) as a LangGraph state machine.
- Integrated OpenAI-compatible LLMs and the Model Context Protocol (MCP) to enable structured tool calling, context-aware reasoning, and seamless collaboration between specialized AI agents.
- Implemented short- and long-term memory using FAISS vector embeddings, improving contextual continuity, reducing redundant retrieval, and enabling more efficient multi-agent task execution.

HR AssistHub — HR Automation Platform | React, FastAPI, LLM, RAG (Team of 4)

- Delivered an end-to-end platform automating employee communication and meeting coordination, replacing a manual workflow of **6 hours/week**.
- Integrated LLM + RAG over internal HR documents for automated email drafting and natural-language meeting scheduling, generating context-aware drafts in under **5 seconds**.

AI-Powered Offline Python Tutor | Ollama (Phi-3), LoRA/QLoRA, React, FastAPI

- Fine-tuned a Phi-3 LLM with **Supervised Fine-Tuning on 10,000+ curated code-explanation pairs**, using LoRA/QLoRA to train on a single 12 GB consumer GPU.
- Improved explanation accuracy by **27%** over the base model on a held-out set of 500 problems, scored by a rubric-based LLM judge.
- Built an offline React + FastAPI interface with real-time error detection, reasoning-based debugging, and personalized feedback.

TECHNICAL SKILLS

Languages: Python, JavaScript

Backend & Web: FastAPI, Node.js, Express.js, RESTful API Design, Distributed Systems, Scalable System Design, Docker, Redis, React.js,

Databases: PostgreSQL, MySQL, MongoDB, pgvector, FAISS, Row-Level Security, Relational Schema Design

AI & ML: LLMs, RAG, LangChain, LangGraph, Model Context Protocol (MCP), BERT Fine-Tuning, LoRA/QLoRA, CNNs, LSTMs, Regression, Clustering

Tools & Practices: Git, GitHub, HuggingFace Transformers, Postman, Agile/Scrum, SDLC, Unit Testing, Code Review

ACHIEVEMENTS

- Solved **400+ Data Structures & Algorithms problems** on LeetCode (arrays, graphs, DP, greedy).
- Runner-Up, Xackthon Hackathon** (2nd of 40 teams); recipient of the **Active Performer Award** for contribution and teamwork, CEG.